

PROJECT README

Student Record Management System

Project Overview

The Student Record Management System is a menu-driven Python application developed to manage student academic records through the command line.

The system allows the user to:

- Add a student
- Search for a student
- Update student marks
- Calculate total marks and percentage
- Display the student result
- Exit the application

The project demonstrates Python functions, dictionaries, lists, loops, conditional statements, and user input.

Features

Add Student: Enter Roll Number, Student Name, and marks of five subjects.

Search Student: Search using the roll number and display stored information.

Update Marks: Update marks of an existing student.

Calculate Total: Calculate total marks of five subjects.

Calculate Percentage: Calculate percentage from subject marks.

Display Result: Display roll number, name, marks, total, percentage, and PASS/FAIL result.

Exit: Close the application.

Technologies Used

- Python 3
- Python Functions
- Dictionary
- List
- For Loop
- While Loop
- If-Else Statements
- Command Line Interface

No external Python libraries are required.

Project Structure

Student-Record-Management-System/

```
■
■■■ student_management.py
■■■ README.md
```

student_management.py: Complete Python source code.

README.md: Project information, setup, and usage instructions.

Requirements

- Python 3.x
- Command Prompt / PowerShell / Terminal

No GUI application or external Python packages are required.

Check Python Installation

Open a terminal or command prompt and run:

```
python --version
```

If needed, try:

```
python3 --version
```

Download or Clone the Repository

```
git clone YOUR_GITHUB_REPOSITORY_URL
cd Student-Record-Management-System
```

Replace YOUR_GITHUB_REPOSITORY_URL with the actual GitHub repository URL.

Environment Setup

No virtual environment or external dependencies are required. The project uses only Python's built-in features.

Dependency Installation

There are no external dependencies. You do not need to run pip install commands.

Configuration

No configuration file or environment variables are required.

Run the Project

```
python student_management.py
```

If your system uses python3:

```
python3 student_management.py
```

```
===== STUDENT RECORD MANAGEMENT SYSTEM =====
```

1. Add Student
2. Search Student
3. Update Marks
4. Display Result
5. Exit

Enter your choice:

How to Use the Program

Step 1: Add a Student — Select 1 and enter roll number, name, and five subject marks.

```
Enter Roll Number: 101
Enter Student Name: Rahul
Enter marks of Subject 1: 80
```

```
Enter marks of Subject 2: 75
Enter marks of Subject 3: 90
Enter marks of Subject 4: 85
Enter marks of Subject 5: 70
Student added successfully!
```

Step 2: Search Student — Select 2 and enter the roll number.

Step 3: Update Marks — Select 3, enter an existing roll number, and enter new marks.

Step 4: Display Result — Select 4 and enter the roll number.

```
----- STUDENT RESULT -----
Roll Number: 101
Name: Rahul
Subject 1 : 80.0
Subject 2 : 75.0
Subject 3 : 90.0
Subject 4 : 85.0
Subject 5 : 70.0
Total Marks: 400.0
Percentage: 80.0 %
Result: PASS
-----
```

Step 5: Exit — Select 5 to terminate the program.

Result Calculation

```
total = sum(marks)
percentage = total / len(marks)
```

Five subjects are assumed, with each subject out of 100, so the average is the percentage.

```
Percentage >= 40  → PASS
Percentage < 40   → FAIL
```

Error Handling

```
Student already exists!
Student not found!
Invalid choice! Please try again.
```

Data Storage

```
students = {}

students = {
    "101": {
        "name": "Rahul",
        "marks": [80, 75, 90, 85, 70]
    }
}
```

Each roll number is used as the dictionary key.

Important Note About Data

This version uses in-memory storage. Student records are lost when the program is closed. No database or external file is required for the current version.

Future Improvements

- Permanent file storage
- Database integration
- Delete student functionality
- Input validation for marks

- Grade calculation
- Display all students
- Student login system
- Graphical User Interface
- Attendance management
- Subject-wise performance analysis

Troubleshooting

Python command not found:

```
python3 --version
```

Install Python 3 and ensure it is added to PATH.

File not found:

```
cd Student-Record-Management-System  
python student_management.py
```

Make sure the source file is named student_management.py.

Project Execution

The complete project can be executed directly from a command-line environment. No GUI, IDE-specific configuration, external libraries, or additional software is required apart from Python 3.

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